Introduce two rate limits: one for the outside and one for the inside. Domains that are at the local server have a different message rate limit. /etc/hosts can be used to decide that.

When contact is installed, it might be installed for different subdomains. Keep a record of where it's installed. Does a subdomain need to be used for separate installations, or can it work in a different folder too?

1. Outside vs. Inside Dual Rate Limits

Two distinct rate limits have been implemented:

- **Outside Limit (Remote Websites)**: Default **10 messages/day** (`rate_limit_outside_per_day`).
- **Inside Limit (Local Server / Subdomains)**: Default **100 messages/day** (`rate_limit_inside_per_day`).

How `/etc/hosts` Determines Inside vs. Outside:

When an incoming message arrives at `index.php`, `is_local_sender()` evaluates the sender:

1. **`/etc/hosts` Lookup**: Checks whether the sender host maps to `127.0.0.1`, loopback (`127.0.0.0/8`, `::1`), or the local server IP (`$_SERVER['SERVER_ADDR']`).
2. **Local Socket Source**: Checks whether the remote connection IP (`$client_ip`) is loopback or local.
3. **Local Suite Registry**: Checks if `/etc/tricurse/<sender_host>/contact.conf` exists on this server.

If any check passes, the sender receives the **Inside** quota. Otherwise, it is constrained by the **Outside** quota. When either limit is exceeded, a `429 RATE_LIMITED_DELAY` status is returned and the dispatch is delayed.

Host Control Panel Settings:

In the Host Control Panel (`index.php`), both quotas can now be customized independently.

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2. Recording Where Contact is Installed

When `install.sh` installs or updates an instance, it registers the installation into a central registry file at:

```
/etc/tricurse/installations.json
```

Each record preserves:

- **Domain / Subdomain**: Hostname (e.g., `client1.example.com` or `example.com`)
- **Web Directory**: The exact directory where files were installed (e.g., `/var/www/html/client1`)
- **Persistent Message Storage**: Directory holding letters outside the web root (`/var/lib/tricurse/contact/<domain>`)
- **Database Path**: SQLite database file (`contact.sqlite`)
- **Configuration File**: `/etc/tricurse/<domain>/contact.conf`
- **Installed Timestamp & User**: Exact installation time and user

Viewing Recorded Installations:

You can review all installations across the machine at any time via:

```
# Using the dedicated CLI tool:
contact_list

# Or directly through the installer:
./install.sh --list
```

---

3. Subdomains vs. Folders for Separate Installations

> **Does a subdomain need to be used for separate installations, or can it work in a different folder too?**

It works in separate folders too. A subdomain is **not required**.

Here is how both approaches compare:

Feature	**Subdomain (`client1.domain.com`)**	**Folder / Subdirectory (`domain.com/client1/`)**
Federated Routing	Delivers to `https://client1.domain.com/index.php`	Delivers to `https://domain.com/client1/index.php` (handled automatically via `$_SERVER['SCRIPT_NAME']`)
DNS & Web Server Setup	Requires DNS A/CNAME records and Nginx/Apache virtual host blocks	**Zero DNS setup required**. Runs directly out of subdirectories under your existing website document root
Loopback `/etc/hosts`	Each subdomain is added to `/etc/hosts` (`127.0.0.1 client1.domain.com`)	The base domain in `/etc/hosts` covers all folders on that machine

Feature	Subdomain (`client1.domain.com`)	Folder / Subdirectory (`domain.com/client1/`)
Cookie & Session Scoping	Browser origins separate cookies automatically	**Path-scoped**: Cookies are scoped to the folder directory (e.g. `path: /client1/`), preventing session collisions between folders
Web Push (PWA)	Distinct origins for Service Worker registrations	Service Workers are scoped to the subpath (`scope: /client1/`), allowing independent push alerts
Database & Config	Isolated per domain in `/var/lib/tricurse/contact/<domain>/`	Isolated per folder in `/var/lib/tricurse/contact/<folder>/`

Recommendation:

- Use **Folders** (`/var/www/html/client1/`, `/var/www/html/client2/`) if you want quick deployments without creating DNS records or configuring Nginx/Apache server blocks.
- Use **Subdomains** (`client1.domain.com`) if you want distinct brand identities and cleaner top-level URLs for each client.